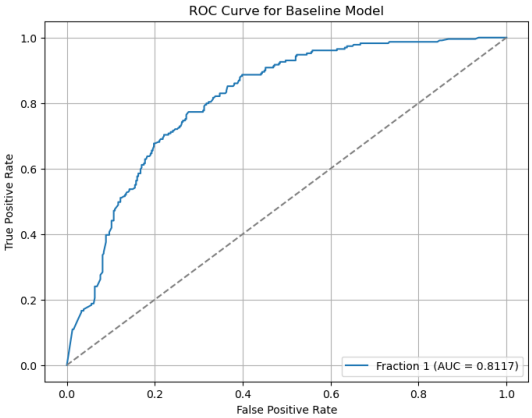
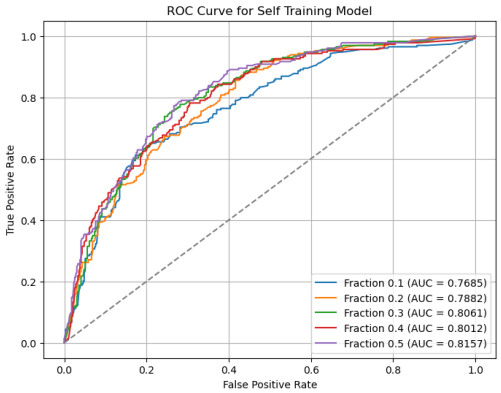
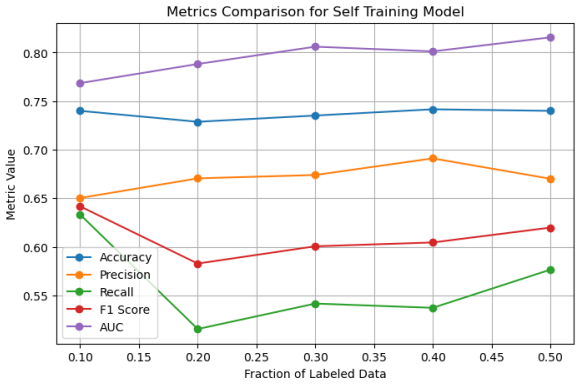
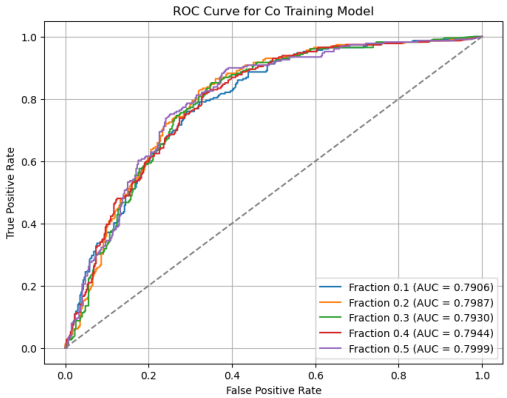
Figures and summary

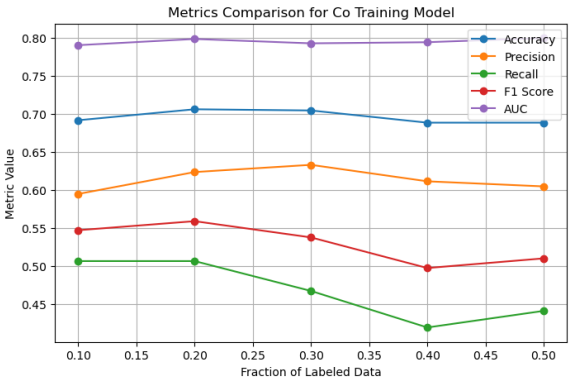
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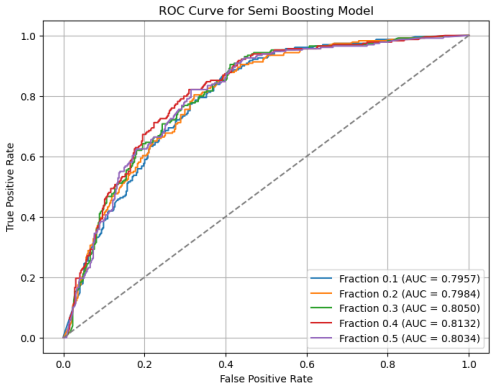


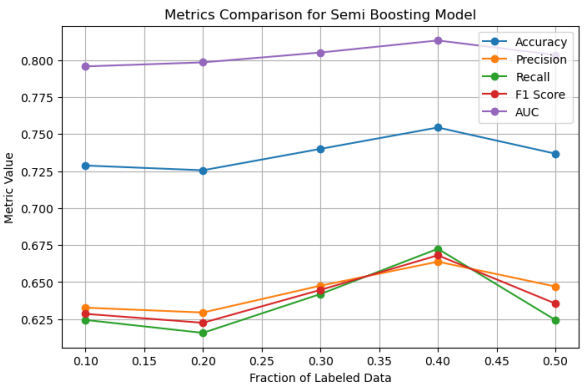


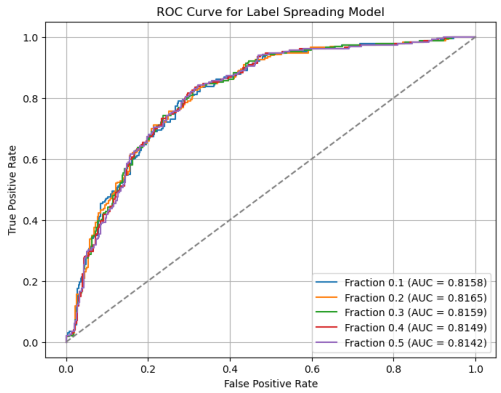


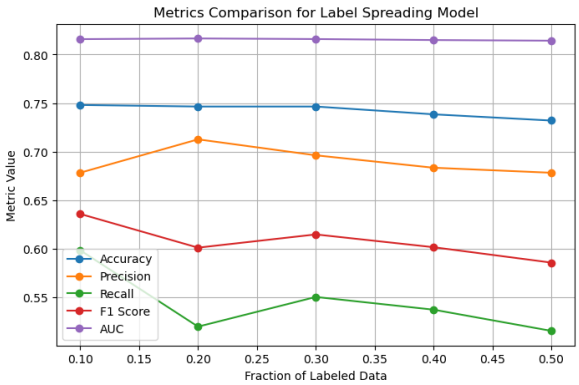


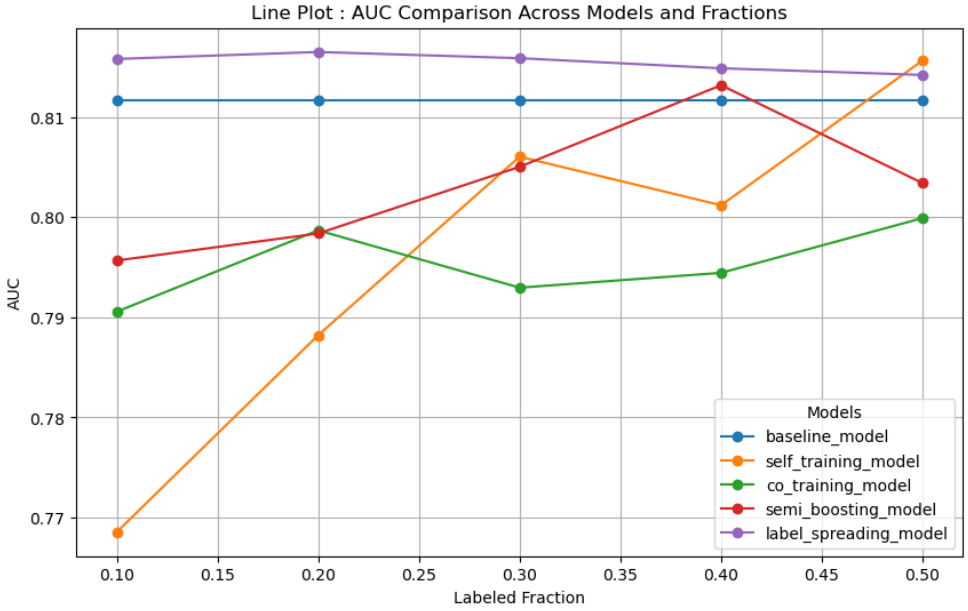


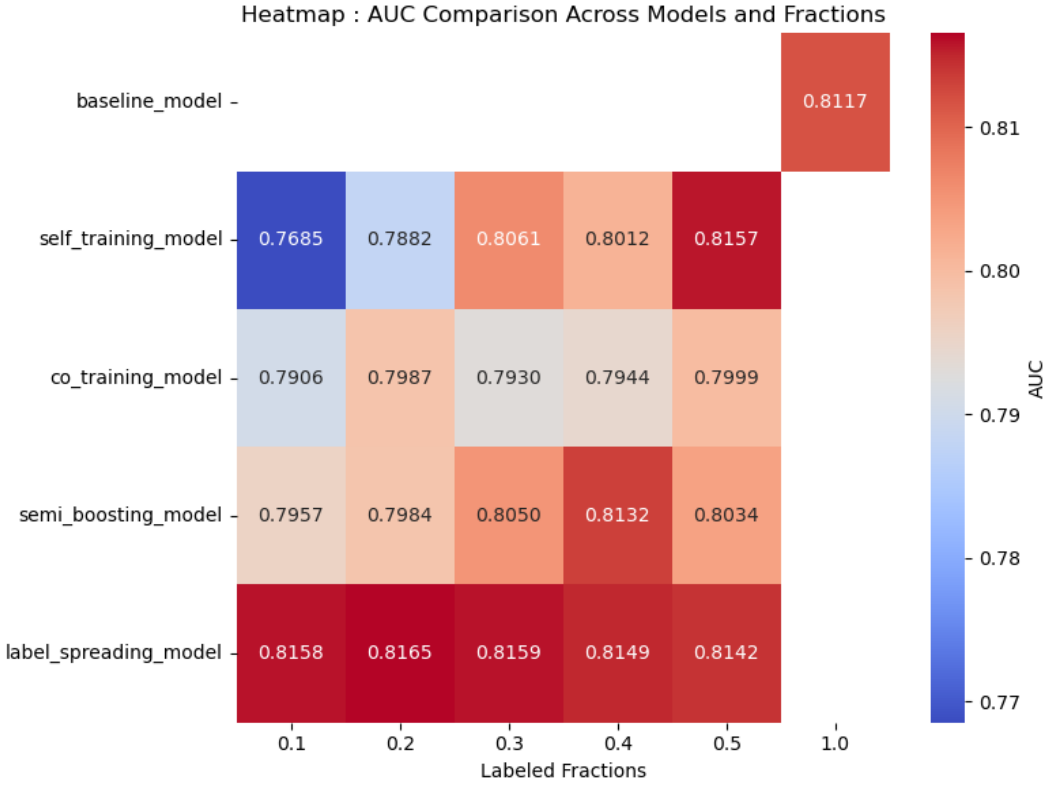


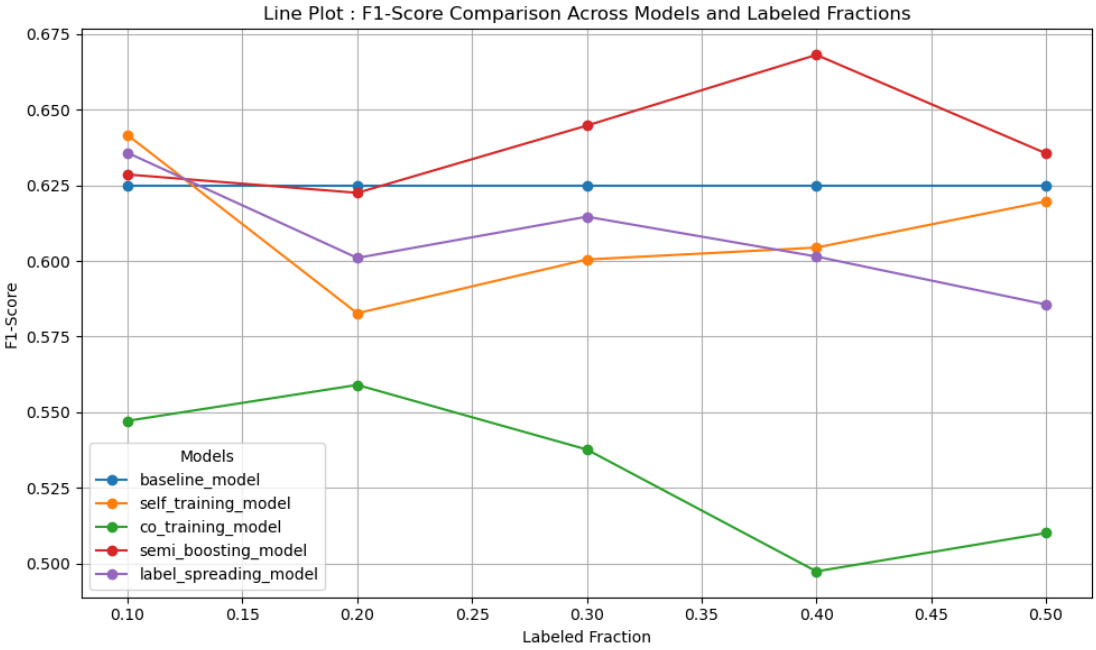


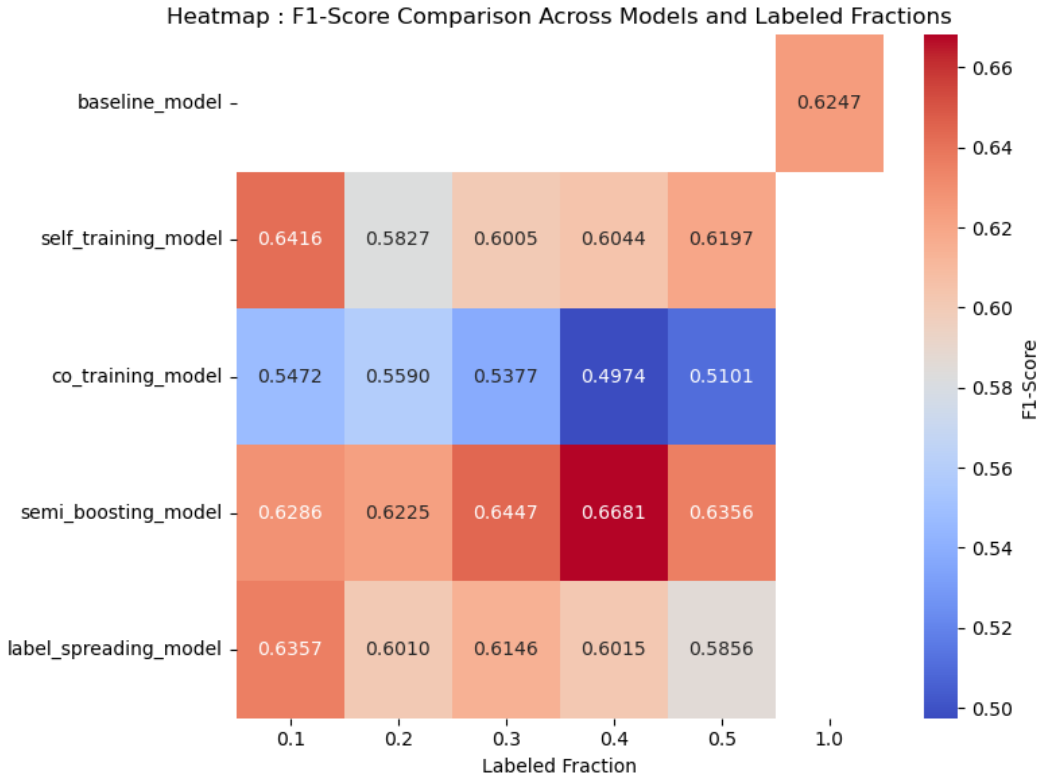

ROC Curve for Label Spreading Model
True Positive Rate
False Positive Rate
Fraction 0.1 (AUC = 0.8158)
Fraction 0.2 (AUC = 0.8165)
Fraction 0.3 (AUC = 0.8159)
Fraction 0.4 (AUC = 0.8149)
Fraction 0.5 (AUC = 0.8142)


Metrics Comparison for Label Spreading Model
Metric Value
Fraction of Labeled Data
Accuracy
Precision
Recall
F1 Score
AUC


Line Plot : AUC Comparison Across Models and Fractions
AUC
Labeled Fraction
Models
baseline_model
self_training_model
co_training_model
semi_boosting_model
label_spreading_model


Heatmap : AUC Comparison Across Models and Fractions
baseline_model
self_training_model
co_training_model
semi_boosting_model
label_spreading_model
0.8117
0.7685 0.7882 0.8061 0.8012 0.8157
0.7906 0.7987 0.7930 0.7944 0.7999
0.7957 0.7984 0.8050 0.8132 0.8034
0.8158 0.8165 0.8159 0.8149 0.8142
Labeled Fractions
AUC





1. The self-training model shows an increasing trend in AUC as the labeled fraction increases. Although the AUC of 0.5 labeled fraction for self-training model with 0.8157 AUC slight surpasses the baseline model (AUC=0.8117), overall AUC of self-training model remain lower than the baseline model. At 0.1 labeled fraction, the F1-score of the self-training model slightly higher than the baseline model, which indicates that self-training model may better generalize at extremely low labeled fractions.

Self-training trained on labeled data predicts pseudo-labels for unlabeled data, these pseudo-labels are then incorporated into the training set iteratively. It's simply to implement, refine predictions and improve performance over multiple iterations theoretically. But the initial model trained on the initial labeled data should be accurate. If the initial model makes incorrect pseudo-labels, these errors can propagate in subsequent iterations.

With fewer labeled data points (10% labeled in this example), the initial model's predictions may introduce noise in pseudo-labels, reducing the model's overall performance. The increased trend of AUC demonstrate its ability to improve as more labeled data become available.

2. Noticeably, co-training model is failed to exceed an AUC of 0.8 or an F1-score of 0.55 across all labeled fractions.

Co-training involves training two models on different views (or subsets) of the data, where each model provides pseudo-labels for the other. This approach is highly effective when the dataset has distinct and complementary feature sets, as the two models can mutually enhance each other's learning. However, in this case, the two views may not be sufficiently distinct or informative, leading to redundancy rather than complementarity in the learning process.

Additionally, co-training is highly sensitive to the quality of pseudo-labels. If the initial model trained on the labeled data is weak, it may generate noisy pseudo-labels. These noisy pseudo-labels can propagate between the two models, compounding errors with each iteration. This error amplification directly impacts precision and recall metrics, leading to a degradation in both F1-scores and AUC.

So, if the dataset lacks clear, distinct feature views or sufficient labeled data to bootstrap robust models, co-training should be avoided.

3. When it comes to semi-boosting, while its AUC is generally lower than the baseline Gradient Boosting model (except at fraction 0.4), the gaps are not significant, and it achieves consistently higher F1-scores (0.6681) across all labeled fractions.

Semi-boosting relies on pseudo-labels generated by an initial model trained on labeled data. If the initial model is weak, the pseudo-labels may be noisy or inaccurate. But boosting prioritizes harder examples during training, the ability to iteratively correct errors make it robust to small amounts of pseudo-label noise. Model becomes better at identifying true positives, even in imbalanced or complex datasets, refine, enhancing recall without overly sacrificing precision. So, semi-supervised model performs good AUC and F1-score.

4. Label spreading model shows minimal variation in AUC, achieves high value outperforming the baseline Gradient Boosting model across all labeled fractions (ranging from 0.8158 to 0.8142). The F1-score declines as the labeled fraction increases, dropping from 0.6357 (fraction 0.1) to 0.5856 (fraction 0.5).

Label spreading leverages the similarity structure of the data by propagating labels through a graph, which ensures that unlabeled samples close to labeled samples in the feature space are correctly classified. In addition, it relies on the feature space structure, the performance does not heavily depend on the labeled fraction. Even with a low labeled fraction (e.g., 0.1), the model can propagate labels effectively, achieving a high AUC.

When the dataset is imbalanced, label spreading is less effective, because it propagates labels based on feature similarity, which can overrepresent majority classes. As more labeled data becomes available, the imbalance is amplified, leading to a drop in recall and, consequently, F1-score.